

Usability of Electronic Health Records by Clinicians in Acute Care: A Scoping Review

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Abstract

Electronic Health Records (EHRs) provide clinicians with real-time digital access to patient health information and documentation. These systems are increasingly adopted worldwide due to their advantages over traditional paper-based records in hospital settings. However, EHRs also present usability challenges that can disrupt clinical workflows. This scoping review examines how EHR usability affects clinicians in acute care environments, with a focus on the factors shaping their experiences. Following the JBI methodology, 11 articles were selected and analyzed using qualitative content analysis. Four key themes emerged: (1) EHR system usability, (2) impact on clinicians, (3) implications for patient safety, and (4) opportunities to enhance usability.

The findings reveal that usability is influenced by factors such as system design and user training, which in turn affect clinician well-being, clinical efficiency, and patient outcomes. While many reports highlight negative experiences with EHR usability, evidence also indicates that well-designed systems can benefit both clinicians and patients. Although this review did not assess whether usability varies by the level of EHR implementation, it identifies this as a valuable direction for future research. Overall, the findings underscore the importance of optimizing system design and fostering collaboration with vendors to ensure EHRs support, rather than hinder, effective clinical practice.

Background

Electronic Health Records (EHRs) are interactive, real-time systems that are designed to provide clinicians with access to review and update patient's health information ([World Health Organization \[WHO\], n.d.](#)). EHRs are interoperable across multiple facilities and offer a digital view of a patient's health history ([Health Information Management, n.d.](#)). Information such as a patient's medical history, diagnoses, treatments, medications, allergies, diagnostic tests, and lab results are accessible within an EHR system ([WHO, n.d.](#)).

By streamlining information management and supporting clinical decision-making, EHRs have transformed healthcare delivery ([Sriram & Subrahmanian, 2020](#)). Benefits such as improved accessibility of information, real-time communication, and comprehensive patient summaries are frequently mentioned within the literature ([Golay et al., 2022](#); [Jedwab et al., 2023](#); [Kaihlainen et al., 2020](#)). These advantages have made EHRs a cornerstone of the healthcare system, driving their increasing adoption worldwide, particularly in hospital settings, as a replacement for paper-based documentation ([Lee et al., 2020](#); [Kaihlainen et al., 2020](#)).

Over the past decade, the prevalence of EHRs has grown rapidly, with high-income countries such as Canada, the United States (U.S.), Sweden, the Netherlands, Japan, and Australia demonstrating steady uptake ([Slawomirski et al., 2023](#)). In the U.S., over 90% of hospitals now use an EHR system ([Kim et al., 2024](#)), while Canada continues its transition, with adoption varying by province and territory ([Lee et al., 2020](#)). In contrast, EHR adoption remains limited in low- and middle-income countries, where financial and infrastructural barriers persist ([Ferry et al., 2021](#)). However, uptake is increasing in these countries too, with now approximately 25% of hospital institutions using an EHR system ([Ferry et al., 2021](#)).

As EHR adoption continues to expand, their role is arguably most critical in acute care settings where the demands of patient acuity are highest and timely task completion is essential (Khariat et al., 2022). In acute care settings, characterized by patients admitted short-term for treatment of acute or critical illnesses or injuries ([Canadian Institute for Health Information, n.d.](#)), clinicians rely on EHRs to help deliver timely and effective care ([Upadhyay & Hu, 2022](#)). However, along with these benefits have come significant challenges. Many EHR systems are designed without fully accounting for the complex workflows and information needs of clinicians ([Lasko et al., 2020](#)). As a result, EHR systems have poor usability because they are not intuitive to use and are not aligned with clinician workflow expectations.

Usability is defined as the degree in which a system is easy to use and the efficiency to perform a task ([Bloom et al., 2021](#); [Center for Quality and Productivity Improvement, n.d.](#); [Shin et al., 2022](#); Shultz & Hand, 2015). In the context of EHRs, it reflects how seamlessly clinicians can navigate and interact with the system and is often measured by the System Usability Scale (SUS) – the industry standard for measuring system usability ([Bloom et al.,](#)

[2021](#)). The SUS ranges from 0 (worst) to 100 (best), with the threshold of acceptable usability at 68 for health technology. A study conducted by [Bloom et al. \(2021\)](#) evaluated the system usability of 25 of the most common EHR systems implemented in emergency departments throughout the United Kingdom. The findings showed that no EHR system met the median industry standard of usability, and five systems scored lower than 50, indicating an unacceptable level of usability.

In acute care, usability takes on a heightened significance, as even [minor inefficiencies \(e.g., disorganized EHR displays or lack of an undo option\)](#) can have a negative impact on clinician workload ([Dunn Lopez et al., 2021](#)). On average, physicians and nurses will spend nearly one-third of their workday on the EHR, which suggests that there is room for functionality improvements and workflow optimization ([Bakhoun et al. 2021](#); [Pinevich et al., 2021](#)). Research has shown that poor usability not only hinders effective navigation of the system but can lead to clinician frustration, increased workload, and patient safety risks ([Upadhyay & Hu, 2022](#)).

While in recent years, researchers across the globe have reviewed the challenges of usability, the focus of these reviews has been on specific elements of EHRs, such as interaction time and specific documentation practices or intervention tools ([Moy et al., 2021](#); [Pinevich et al., 2021](#)). Beyond the usability of isolated functionality, limited attention has been given to the broader influence of the entire EHR as a complete, integrated system within the context of acute care. The Open Science Framework was reviewed, and to the best of the author's knowledge, no in-progress or published scoping reviews address this topic. Therefore, this gap highlights the need to consolidate evidence on EHR use in acute care to help guide improvements in system design and usage in this area. The objective of this scoping review is to examine existing research on EHR usability and its effect on clinicians. Specifically, it aims to:

1. Understand how the usability of an EHR system affects clinicians within acute care;
2. Identify design features of EHR systems and highlight challenges, usability improvements, and system enhancements; and
3. Identify gaps in the current literature to inform future research.

Review Question

What is known about the implications of EHR usability for clinicians in acute care settings?

Methods

The JBI methodology for scoping reviews ([Peters et al., 2024](#)) was used to conduct this review. A scoping review was chosen because it aligns with the review's objectives and is well-suited to synthesizing and mapping evidence on broad, complex topics such as EHR

usability ([Aromataris et al., 2024](#)). The results are reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews (PRISMA-ScR) ([Tricco et al., 2018](#)).

Inclusion Criteria

Participants

Studies focused on the usability outcomes of health care providers and/or their student role equivalents were considered for this review. A health care provider refers to a licensed medical personnel or clinician who provides health care services within the scope of their practice ([National Cancer Institute, n.d.](#); [University of California Berkeley; n.d.](#)). Common types of health care providers include physicians, nurses, pharmacists, and physiotherapists. Similarly, their student role equivalents include, but are not limited to, medical students, residents, and nursing students. Studies focusing on non-licensed health care providers were excluded.

Concept

Studies that examined the usability of EHRs by clinicians, as well as those exploring clinician experiences with EHRs were considered for this review. Included studies focused either on the usability of the EHR as a whole – examining how clinicians engage with the full range of system functions – or discussed the usability of core system attributes that are common across different EHRs, such as flowsheets, medication administration records, electronic order entry, care plans, documents, and Admission-Discharge-Transfer (ADT) navigations ([Cho et al., 2024](#)). Studies that focused solely on the design, testing, or usability of a specific tool integrated into an EHR (e.g., clinical decision support systems, medication alerts) were excluded. Since the terms electronic health records (EHRs) and electronic medical records (EMRs) are used interchangeably within the literature ([Shin et al., 2022](#)) either term was accepted within this review.

Context

This review included studies situated in any geographic location, regardless of country of origin or sociocultural setting. The focus was on studies that evaluated the usability of EHRs for clinicians working within an acute care setting such as inpatient units, emergency departments, or critical care. Studies conducted in outpatient clinics, ambulatory care settings, primary care, or in hospital-based facilities that did not use an EHR were excluded.

Types of Sources

Published primary research studies (of any type) were considered for inclusion. Scoping reviews, systematic reviews, and meta-analyses were not included. For the context of providing evidence-based information for health and research, peer-reviewed evidence is considered the gold standard ([Maramba et al., 2019](#)). Therefore, only peer-reviewed published literature in the English language was included in this review.

Search Strategy

The search strategy aimed to locate published studies using a two-step approach. First, an initial limited search of MEDLINE and CINAHL was completed to identify relevant articles. The text words contained in the titles and abstracts of these articles, along with the medical subject headings (MeSH) were used to develop a full search strategy. The Yale MeSH Analyzer ([Grossetta Nardini & Wang, 2025](#)) was also used to identify any additional subject terms/MeSH that had not already been identified. As scoping reviews are an iterative process ([Peters et al., 2024](#)), additional keywords were incorporated into the search strategy and further refined with consultation from a health sciences librarian. Finally, the search strategy was translated and tailored to each database based on the identified keywords and MeSH.

Due to time and resource constraints, the search was limited to two databases: MEDLINE (Ovid) and CINAHL (EBSCO). These databases were strategically selected as they are both leading databases for biomedical and health sciences research and provide comprehensive peer-reviewed medical literature ([EBSCO Information Services, n.d.-a](#); [EBSCO Information Services, n.d.-b](#)). These databases were searched for published studies from January 1, 2020, to January 14, 2025. As technologies have advanced rapidly within the health care system ([Junaid et al., 2022](#)), only the most recent five years were considered for this review. A detailed search strategy for each of the two databases can be found in Tables 1 and 2.

Table 1

Search Strategy for MEDLINE

[Sheryl-Drews-Table1Download](#)

Table 2

Search Strategy for CINAHL

[Sheryl-Drews-Table2Download](#)

Study Selection

Following the search, all identified citations were collated and uploaded into Rayyan software ([Ouzzani et al., 2016](#)) and the duplicates were removed. The title and abstracts of the remaining articles were screened by the author for relevance to the research question. Thereafter, the full-text of studies that potentially met the inclusion criteria were retrieved. The full-text screening was performed independently by the primary author of this review. During this stage in the study selection process, two articles were discussed with the author's supervisor and project committee member to determine inclusion. The results of the search, along with the reasons for excluding full-text articles, were reported in full and presented in a PRISMA flow diagram ([Page et al., 2021](#)).

Data Extraction

The primary author of this review independently extracted data from the included articles and summarized this information into a Microsoft Excel data extraction worksheet developed for this project. Information captured within this worksheet was based off the suggestion from the JBI methodology ([Peters et al., 2024](#)) and modified to include information related to the scoping review question. These details include the study author(s), publication year, country of origin, study methodology, study objective/aims, participants, and key findings relevant to the research question.

Data Analysis and Presentation

According to [Peters et al. \(2024\)](#), scoping reviews should not perform a thematic analysis/synthesis but can instead provide an in-depth analysis using descriptive qualitative content. Therefore, a qualitative content analysis as outlined by [Elo and Kyngäs \(2008\)](#) was followed for this scoping review using the outlined three steps: 1) preparation; 2) organizing and; 3) reporting. During the preparation phase, the data of the selected studies (n=11) was reviewed by the primary author to become familiar with the content and information. Next, during the organizing phase, the data was condensed into codes and similar codes were grouped together to form sub-categories/sub-themes ([Elo & Kyngäs, 2008](#)). A Microsoft Excel worksheet was used to document the coded data. Finally, during the reporting step, the analysis results were reported through a narrative summary which describes how the results relate to the review's objectives and question. The narrative summary is outlined in the results section below.

Results

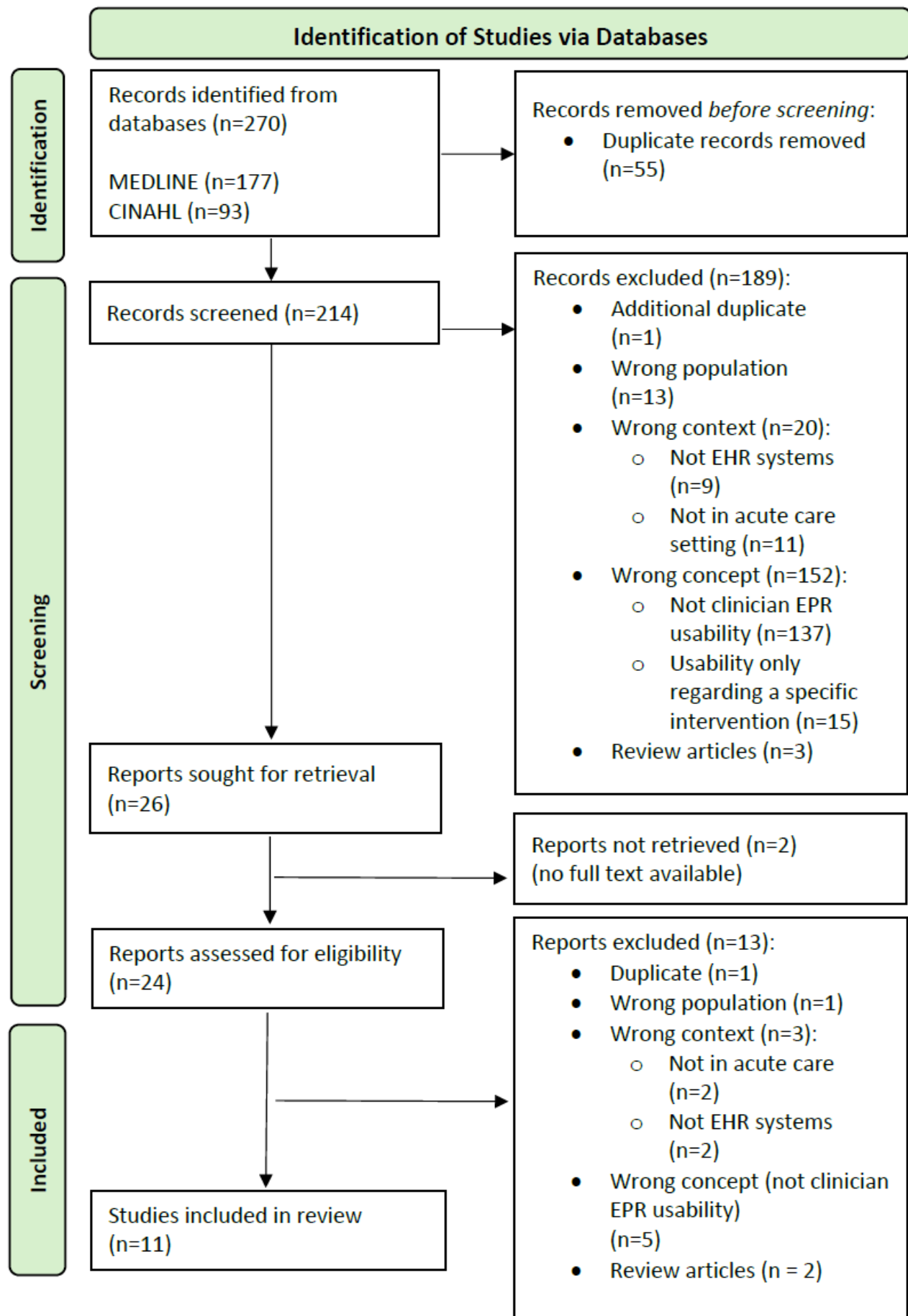
Study Inclusion

Through the database searching, 270 studies were identified for this scoping review. Of these, 55 studies were excluded due to duplication. The remaining 214 studies were screened for inclusion criteria, and 189 were excluded for not meeting the criteria. A total of 26 studies were selected for full-text screening; however, two were excluded because the full text was not accessible.

The remaining 24 articles were read in full, and 13 studies were excluded for the following reasons: duplicate article (n=1); wrong population (n=1); context not within acute care (n=2); context not related to EHR systems (n=2); and the study addressed the wrong concept (not clinician EHR usability) (n=5). Refer to Figure 1 for the full list of reasons for study exclusion. A total of 11 articles were included in this review.

Figure 1

Screening Process of Selected Articles



Characteristics of Included Studies

The studies selected for this review (n=11) were primarily mixed-method studies (n=6), followed by quantitative (n=3) and qualitative (n=2) studies (see Table 3). The mixed methods combined qualitative interviews ([Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#); Ross, 2020; [Shin et al., 2022](#)) and debriefing questions/questionnaire ([Alami et al., 2022](#)) with quantitative measures such as time spent on the EHR ([Alami et al., 2022](#); [Cho et al., 2024](#); Ross, 2020) and Likert Scale surveys ([Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#); [Shin et al., 2022](#)). The quantitative studies used validated surveys ([Classen et al., 2023](#); Khairat et al., 2022; [Kutney-Lee et al., 2021](#)), usability questionnaires ([Classen et al., 2023](#)), objective pupil measurements (Khairat et al., 2022), and data from hospital databases and abstracts ([Kutney-Lee et al., 2021](#)). The qualitative studies used interviews ([Lasko et al., 2020](#); [Senathirajah et al., 2022](#)), think aloud usability testing ([Senathirajah et al., 2022](#)), and collected data through direct observations ([Lasko et al., 2020](#)).

Table 3

Study Characteristics

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Of the 11 studies included, the majority originated in the U.S. (n=8), followed by South Korea (n=1), Guatemala (n=1), and Australia (n=1). All studies evaluated the usability/use of EHRs by clinicians. Most of the studies (n=5) aimed to examine the clinicians' experience of using EHRs ([Cho et al., 2024](#); [Classen et al., 2023](#); [Lasko et al., 2020](#); Khairat et al., 2022; [Kutney-Lee et al., 2021](#)), while the remaining (n=6) explored usability problems and benefits ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)) and the clinicians' perceptions of usability ([Mullins et al., 2021](#); Ross, 2020).

Clinician populations varied across studies, including those working in critical care ([Cho et al., 2024](#); Khairat et al., 2022), inpatient units ([Kutney-Lee et al., 2021](#)) such as: adult medical-surgical, pediatrics, and acute care neurology ([Alami et al., 2022](#); [Cho et al., 2024](#); [Senathirajah et al., 2022](#)), and the Emergency Department ([Aldredge et al., 2020](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); [Shin et al., 2022](#)). Two studies did not specify the unit/department ([Classen et al., 2023](#); Ross, 2020). The studies included various health care providers such as nurses, physicians, physician extenders (e.g., physician assistants), pharmacists, and medical students. No studies mentioned students from other healthcare disciplines. Some studies focused on a single provider type while other studies included a combination (see Table 4).

Table 4

Clinician Populations Represented in Included Studies

Population	Number of Studies
Nurses	4
Physicians	3
Physicians and Nurses	1
Physicians and Pharmacists	1
Physicians and Medical Students	1
Physicians and Physician Extenders	1

The sample sizes of the studies ranged from six participants ([Senathirajah et al., 2022](#)) to over 12,004 ([Kutney-Lee et al., 2021](#)). Of the included studies, five reported the clinicians' EHR experience, with all participants having prior experience and varying levels of proficiency. Seven studies did not specify the clinicians' experience with EHRs (Table 5).

Table 5

Study Context and Population

[Sheryl-Drews-Table5Download](#)

Review Findings

From the content analysis using the framework by [Elo and Kyngäs \(2008\)](#), four overarching themes were identified with regards to the usability of EHRs and the implications for clinicians. The first theme, *usability of EHR systems*, describes how design, functionality, and access to the system influence usability for clinicians. The second theme, *effects on clinicians*, describes how system usability affects clinician well-being and their clinical practice. The third theme, *effects on patient safety*, outlines how EHR usability by clinicians can impact patient outcomes. Finally, the fourth theme, *recommendations to improve usability*, identifies ways to enhance the use of the system. This is summarized for each of the 11 studies in Table 6.

Table 6

Usability of EHR Systems

Functionality

Six studies identified that the lack of essential features within the EHR system can affect clinicians' ability to efficiently complete tasks ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Kutney-Lee et al., 2021](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Various examples were provided of features that were missing from EHR systems. These included features to assist with searching for information ([Cho et al., 2024](#); [Senathirajah et al., 2022](#)), functionality that allowed for quick 'undo' or deletions ([Cho et al., 2024](#)), automatic calculators ([Alami et al., 2022](#)), and integration with other systems ([Aldredge et al., 2020](#)).

Design/Layout

A consistent theme across all eleven articles was that the design and layout of the EHR system either supported or hindered clinicians' ability to efficiently retrieve and document patient information ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Khairat et al., 2022](#); [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); [Ross, 2020](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Five studies identified elements that improved efficiency, such as designs that facilitated quick data search and entry (e.g., displays that minimize unnecessary detail) ([Aldredge et al., 2020](#); [Lasko et al., 2020](#)), the ability to identify patient trends and risks (e.g., within lab results) ([Ross, 2020](#); [Shin et al., 2022](#)), increased documentation legibility ([Aldredge et al., 2020](#)), and customization options that allowed clinicians to prioritize their workflow ([Ross, 2020](#)).

On the contrary, nine studies highlighted inefficiencies caused by the EHR design, which negatively influenced clinician performance ([Alami et al., 2022](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Khairat et al., 2022](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); [Ross, 2020](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Among these, three studies identified specific formatting challenges that made information difficult to locate due to hidden or poorly structured content, small font sizes, and the inability to adjust formatting based on individual preferences ([Alami et al., 2022](#); [Cho et al., 2024](#); [Shin et al., 2022](#)). Additionally, seven studies identified problems with the EHR screen/display design, including busy screens ([Senathirajah et al., 2022](#); [Shin et al., 2022](#)), information overload ([Alami et al., 2022](#); [Khairat et al., 2022](#); [Mullins et al., 2021](#)), and layouts that did not align with clinician workflow ([Alami et al., 2022](#); [Cho et al., 2024](#)). These issues have been reported to compromise the emotional well-being of clinicians through careless clicking and additional time spent trying to navigate to find information within the EHR ([Alami et al., 2022](#); [Classen et al., 2023](#); [Cho et](#)

[al., 2024](#); Khairat et al., 2022; [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); [Shin et al., 2022](#)). In addition to navigation challenges, duplicated documentation and redundancies were reported by three studies, which created inefficient documentation practices for clinicians ([Cho et al., 2024](#); Ross, 2020; [Senathirajah et al., 2022](#)).

EHR Access

Three studies identified that the ability to access the EHR impacted reliability (e.g., system outage) and usability for clinicians ([Aldredge et al., 2020](#); [Mullins et al., 2021](#); [Shin et al., 2022](#)). Barriers to access create usability challenges, with some clinicians reporting that they eventually “gave up” on using the system ([Mullins et al., 2021](#)). Identified issues included system outages ([Aldredge et al., 2020](#)), poor system performance ([Shin et al., 2022](#)), login difficulties, and inadequate infrastructure, such as an insufficient number of computers available to complete documentation ([Mullins et al., 2021](#)).

Effects on Clinicians

Clinician Well-being

The usability of an EHR system and its relationship to clinician well-being was a common theme identified within nine studies ([Alami et al., 2022](#); [Cho et al., 2024](#); [Classen et al., 2023](#); Khairat et al., 2022; [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); Ross, 2020; [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Four studies reported positive outcomes, indicating that as clinicians gained more experience and spent more time using the system, their workload decreased, and both usability and satisfaction improved (Khairat et al., 2022; [Mullins et al., 2021](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). However, the majority of studies identified poor usability as a contributor to negative outcomes. Inefficient EHR systems increased stress ([Classen et al., 2023](#)), frustration ([Classen et al., 2023](#); [Lasko et al., 2020](#)), and in extreme cases, strong feelings of dislike or even hatred towards the system ([Senathirajah et al., 2022](#)). Three studies identified that poor EHR usability increased cognitive workload. Clinicians had to navigate complex systems, which required significant memory demand to locate and recall patient information from different areas of the system ([Alami et al., 2022](#); Khairat et al., 2022; [Lasko et al., 2020](#)). In addition, the extra time required to use a system that was not user-friendly contributed to increased clinician workload, burden ([Cho et al., 2024](#); [Lasko et al., 2020](#); Ross, 2020), and fatigue ([Classen et al., 2023](#); Khairat et al., 2022). One study further expanded on these findings, noting that poor usability ultimately led to job dissatisfaction, intention to leave, and clinician burnout ([Kutney-Lee et al., 2021](#)).

Clinical Practice

Six studies identified implications of EHR usability on clinical practice. One study found that EHRs improved diagnostic accuracy and supported clinical decision making ([Mullins et al., 2021](#)). However, the remaining five reported negative outcomes. Poor usability often increased the time clinicians spent away from their patients (Ross, 2020; [Shin et al., 2022](#)). One study reported that as much as 50% of a clinician's shift was spent on the EHR (Ross, 2020), which reduced direct patient interaction ([Lasko et al., 2020](#)). In addition, poor usability was found to increase errors (e.g. medication errors) made by clinicians ([Classen et al., 2023](#); [Shin et al., 2022](#)). Specifically, ineffective displays and system features that are difficult to use led to an increase in accidental errors ([Shin et al., 2022](#)).

Effects on Patient Safety

More than half of the studies (n = 9) identified a relationship between EHR usability and the impact on patient safety and patient care ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Kutney-Lee et al., 2021](#); [Mullins et al., 2021](#); Ross, 2020; [Shin et al., 2022](#)). Well-designed systems improved patient safety and quality of care by supporting clinical decision making through easily identifying patient information and providing decision support (e.g., medication interactions) ([Classen et al., 2023](#); [Mullins et al., 2021](#); Ross, 2020). However, when EHR usability was poor, clinicians developed workarounds to compensate for system limitations, which introduced new safety risks, delayed treatment, and increased likelihood of medication errors ([Alami et al., 2022](#); [Kutney-Lee et al., 2021](#); Ross, 2020). One study specifically linked poor usability to worsened patient outcomes including higher patient mortality, increased re-admission rates, and a greater likelihood of inpatient death (Kutney et al., 2021).

Recommendations to Improve Usability

All 11 studies provided recommendations on how to improve the usability of EHRs for clinicians ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); Khairat et al., 2022; [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); Ross, 2020; [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). These suggestions are categorized into functionality improvements, design improvements, and EHR training (e.g., continuous training opportunities).

Functionality Improvements

Five studies identified opportunities to enhance basic EHR functionality to improve clinician efficiency and reduce burden. System integration with other technology platforms was emphasized as a key area for improvement, with three studies advocating for better interoperability between different applications to streamline workflows ([Cho et al., 2024](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Filtering and searching functions were also

identified as an area needing improvement, with recommendations for search bars ([Cho et al., 2024](#)), filtering options ([Lasko et al., 2020](#); [Shin et al., 2022](#)), and a single-scroll feature for navigating complex visual elements ([Shin et al., 2022](#)).

Automation was another common recommendation, with two studies supporting features such as automatic task reminders and predictive tools based on the patient's condition ([Cho et al., 2024](#); [Senathirajah et al., 2022](#)), autocomplete features for documentation ([Shin et al., 2022](#)) and automatic calculations for a patient's fluid intake ([Alami et al., 2022](#)). Visual notifications were also suggested to improve usability, with four studies recommending real-time alerts and reminders to help clinicians track patient statuses ([Alami et al., 2022](#); [Cho et al., 2024](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). However, two studies caution that too many alerts can cause alert fatigue resulting in desensitization to these prompts ([Classen et al., 2023](#); [Shin et al., 2022](#)). Therefore, while alerts are recommended by some studies, one study recommends minimizing the use of them and being more intentional with the alerts that are used ([Shin et al., 2022](#)).

Design Improvements

Eight studies suggested opportunities to improve the EHR design to not only maximize the time spent with patients but to better support clinicians ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Lasko et al., 2020](#); [Kutney-Lee et al., 2021](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). A key concern was the documentation burden, with several studies recommending standardization of note templates and elimination of redundancy through design improvements ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). Enhancing the EHR display was another common theme, especially to minimize time spent searching for relevant data ([Lasko et al., 2020](#)) and to improve navigation of the system ([Cho et al., 2024](#); [Shin et al., 2022](#)). Recommendations included improving information visibility ([Shin et al., 2022](#)), standardizing the presentation of information ([Kutney-Lee et al., 2021](#)), and offering a flexible display that clinicians can adjust as needed ([Lasko et al., 2020](#); [Senathirajah et al., 2022](#)). In addition, user-centered design and system personalization were also suggested as ways to reduce clinician burden, including design elements that allow users to tailor the system to their needs ([Cho et al., 2024](#); [Senathirajah et al., 2022](#)).

EHR Training

Three studies reported that clinicians received insufficient training and do not have the appropriate knowledge to troubleshoot technology problems ([Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#)). Two studies also found that clinicians had a low perception of the system's value, which has been identified as a barrier to using the system ([Cho et al., 2024](#); [Mullins et al., 2021](#)). To address this, five studies recommended offering training on an on-going basis (rather than a one-time session), to not only improve usability but to promote

the importance and the value of using EHRs ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#); Ross, 2020). When training clinicians to use the EHR, instruction should align with the workflow of the clinician's department or unit ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2020](#); [Kutney-Lee et al., 2021](#); [Senathirajah et al., 2022](#)).

Discussion

Implications of EHR Usability

This scoping review identified 11 studies examining clinician use of EHRs in acute care settings, highlighting a notable scarcity of research in this area. Despite the limited research, the findings provide valuable insight into how EHR usability influences both clinicians and their interactions with patients. Overall, this review demonstrated that clinicians perceive EHR systems as presenting more challenges than benefits. While EHRs have shown improvements in communication, workflow, and clinician efficiency ([Upadhyay & Hu, 2022](#)), only five studies confirmed that EHRs assisted with clinical activities ([Aldredge et al., 2020](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); Ross, 2020; [Shin et al., 2022](#)). These benefits were attributed to improved diagnostic accuracy through real-time access to information such as lab results and increased legibility. However, the limited number of studies demonstrating clear benefits suggests that, despite the intended advantages of EHRs, their use in acute care settings is more often hindered by poor usability due to design flaws, missing functionality, and poor workflow integration. These challenges indicate that the practical experience of using EHRs does not adequately align with frontline clinical needs, leading to frustration and inefficiencies rather than improved care delivery.

When EHRs do not function as intended in real-world clinical settings, they become obstacles rather than valuable tools. Instead of enhancing care delivery, poorly designed systems introduce additional workload and burden to clinicians ([Budd, 2023](#); [Cho et al., 2024](#); Ross, 2020; [Shin et al., 2022](#)), ultimately undermining the very goal of implementing an EHR system in the first place. These challenges, in turn, negatively affect clinician well-being and clinical practice overall ([Alami et al., 2022](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Lasko et al., 2020](#); [Khairat et al., 2022](#); [Kutney-Lee et al., 2021](#); [Mullins et al., 2021](#); Ross, 2020; [Senathirajah et al., 2022](#); [Shin et al., 2022](#)).

Although this review focused on the implications for clinicians, it is also important to recognize the broader impact of EHR usability on patient care. Clinicians, driven by their moral and ethical responsibilities, are committed to providing safe and high-quality care ([Connor et al., 2023](#)). Their commitment extends to continually seeking ways to improve patient outcomes ([Carney, 2006](#)), making it necessary to consider the broader effects of EHR usability. Poor usability leads to inefficiencies and user challenges that not only disrupt clinical workflows but can also result in mistakes that compromise patient safety ([Alami et al.,](#)

[2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Kutney-Lee et al., 2021](#); [Mullins et al., 2021](#); Ross, 2020; [Shin et al., 2022](#)). To address these system limitations, clinicians develop workarounds, which can lead to additional errors ([Alami et al., 2022](#); Ross, 2020). Moreover, the extra time required to navigate a poorly designed system takes time away from patient care (Ross, 2020; [Shin et al., 2022](#)).

Overall, the findings from this review emphasize that when EHRs are not designed with usability in mind, they affect not just clinicians but also the care they provide to patients. Despite the potential benefits of EHR systems, only four studies demonstrated that well-designed EHR systems could improve patient outcomes ([Aldredge et al., 2020](#); [Classen et al., 2023](#); [Mullins et al., 2021](#); Ross, 2020). The imbalance of negative to positive outcomes suggests that EHR systems have yet to realize their full potential in acute care settings. Therefore, prioritizing usability improvements is essential not only for clinician efficiency and satisfaction but also for ensuring that EHRs fulfill their intended role in enhancing patient care ([Classen et al., 2023](#); [Shin et al., 2022](#)).

Improving EHR Usability

Research suggests that EHR usability issues often arise because these systems are typically developed by non-clinical organizations that do not account for hospital workflows ([Hettinger et al., 2021](#)). As a result, this misalignment complicates clinical practice, with clinicians historically feeling excluded from health system initiatives, resulting in designs that fail to prioritize care coordination/workflow and clinical decision-making ([Bakhoun et al., 2021](#)). Consequently, these systems become difficult to use in practice, as highlighted by [Cho et al. \(2024\)](#) and [Khairat et al. \(2022\)](#).

To address this, one study in this review advocated for the inclusion of clinicians in the EHR development process to help bridge this gap ([Kutney-Lee et al., 2021](#)). By involving clinicians in these discussions, system designs are more likely to reflect the practical needs of those using them. Additionally, it is important to recognize that different providers (physicians, nurses, and pharmacists, etc.) have varying needs and priorities regarding system functionality ([Jung et al., 2021](#)). The difference in perspectives is highlighted in the study by [Mullins et al. \(2021\)](#), where 91% of pharmacists agreed that the EHR was presented in a user-friendly way, whereas only 44% of physicians shared that view. Therefore, incorporating diverse perspectives into the design process is crucial to ensuring the system works effectively across various roles.

Beyond usability, clinician involvement is also critical to avoid introducing unintended task burdens. Additional studies report that EHR implementations have introduced extra clerical tasks for clinicians (e.g., entering diagnoses, orders, and notes) that exceed the traditional responsibilities with paper-based records ([Budd, 2023](#); [Shanafelt et al., 2016](#)). These additional tasks can be cumbersome and detract from patient care, thereby increasing clinician workload ([Shanafelt et al., 2016](#)). [Shin et al. \(2022\)](#) emphasized similar challenges

that often emerge after EHR adoption, particularly cumbersome documentation practices ([Cho et al., 2024](#); Ross, 2020; [Senathirajah et al., 2022](#)). Consequently, this reinforces the importance of clinicians participating in both design and implementation conversations to ensure that the EHR is not only easy to use but also has clearly defined roles and responsibilities.

While clinician involvement is critical, responsibility for usability does not rely solely on healthcare organizations or end users/clinicians. [Cho et al. \(2024\)](#) reported that at baseline, EHR components are not designed to meet the standard of technology usability. These findings raise important questions: if EHR systems do not meet minimum usability upon implementation, should vendors bear greater responsibility for ensuring their products meet at least these standards, and should the government play a stronger role in enforcing usability standards to ensure that vendor contracts prioritize safe and reliable systems ([Institute of Medicine, 2012](#))? While [Classen et al. \(2023\)](#) recognized that vendors should take more responsibility, the findings of this review make it clear that issues with EHR usability arise from many different causes, such as design shortcomings, poor integration into clinical workflows, insufficient training, and poor system reliability ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Khairat et al. 2022](#); [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); [Mullins et al., 2021](#); [Senathirajah et al., 2022](#)). As these challenges are multi-faceted, improving usability requires a collective approach and cannot fall on one group alone. For this reason, it should be a shared responsibility to improve EHR usability ([Classen et al., 2023](#)). For example, the vendor can be responsible for ensuring that the EHR design and functionality are available and work appropriately, while the healthcare organization ensures that the computers are functional and the software is up to date to ensure fast and reliable EHR response times ([Sittig et al., 2018](#)).

As a starting point to mitigate the usability challenges, hospital organizations and EHR vendors should prioritize improving the usability of EHRs by incorporating workflow considerations into both system enhancements and clinical training ([Alami et al., 2022](#); [Kutney-Lee et al., 2021](#); [Senathirajah et al., 2022](#)). Most studies provided various suggestions to enhance usability, focusing on EHR design changes and additional functionality ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Classen et al., 2023](#); [Kutney-Lee et al., 2021](#); [Lasko et al., 2020](#); [Senathirajah et al., 2022](#); [Shin et al., 2022](#)). In this context, design changes need to balance the standardization of information ([Shin et al., 2022](#); [Kutney-Lee et al. 2021](#)) with the flexibility to offer customization based on clinician preferences ([Lasko et al., 2020](#); [Senathirajah et al., 2022](#)). These goals often contradict each other, and finding a balance between them is difficult ([Bloom et al., 2021](#)), as it requires aligning the needs of patients, clinicians, and data reporting ([Sinsky et al., 2021](#)).

Several studies also identified insufficient training as a significant knowledge gap ([Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#)), which has been linked to poor usability, low acceptability, and distrust towards the system ([Lasko et al., 2020](#); [Jung et al., 2021](#)).

Offering routine education for clinicians was frequently recommended and is considered an essential starting point for the successful implementation of EHR systems ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Mullins et al., 2021](#)) and a facilitator for EHR adoption ([Jung et al., 2021](#)). However, while the type of training, whether virtual or hands-on, was not specified, research consistently supports the value of repetitive training in promoting user acceptance and usability ([Alexiuk et al., 2024](#); [Jung et al., 2021](#); [Upadhyay and Hu, 2022](#)).

Gaps

One study in this review ([Kutney-Lee et al., 2021](#)) suggested that more comprehensive EHR systems, with a broader range of features, tend to have better usability. In contrast, four studies focused on core EHR functions, which were often criticized for lacking essential features that should already be present ([Alami et al., 2022](#); [Aldredge et al., 2020](#); [Cho et al., 2024](#); [Senathirajah et al., 2022](#)). While the idea of a feature-rich system was not widely discussed, it's unclear whether adding more features will improve usability or further complicate clinicians' experience. While this review focused on core features necessary for clinical practice, the impact of additional software features or systems with full functionality verses those with only the necessary amount, could be a valuable area for future research.

Additionally, one study noted that its Emergency Department, while using an EHR, still relied on paper for certain tasks, such as physician orders ([Aldredge et al., 2020](#)). In contrast, another study described that its facility was "more electronic," with physician orders entered directly into the EHR system ([Cho et al., 2024](#)). The results in this review did not determine how varying levels of implementation (e.g., fully verses partially electronic facilities) influenced usability. However, external research suggests that usability may be linked to the healthcare organization and its level of EHR implementation ([Bloom et al., 2021](#)). Whether this impacts clinicians' experiences remain unclear, highlighting the need for further research in this area.

Limitations

As with any review, these results should be interpreted with the consideration of some limitations. While this scoping review is extensive and included a comprehensive search strategy based on the JBI methodology ([Peters et al., 2024](#)), only two databases were included due to time constraints. As a result, this may have led to the omission of relevant studies. Second, only studies published in English were included as the authors did not have access to translation services. Future studies should be inclusive of all languages to provide a more wholesome understanding of EHR usability. Lastly, a limitation of this review is the lack of international comparison. Over 70% of the included studies originated in the U.S.

Although efforts were made to include studies from various countries, this lack of geographical diversity may limit the context and relevance to other healthcare systems, as countries vary in terms of resources and infrastructure ([Thomas, 2023](#)).

Conclusion

The usability of EHR systems within acute care settings plays a critical role in shaping clinicians' experiences and patient outcomes alike. The findings of this review provide a better understanding of current EHR system limitations, the risks involved, and opportunities for improvement. A key takeaway is the significant relationship between usability, clinician well-being, clinical practice, and patient safety. While the overall usability of EHRs has been largely negative, there is evidence that when designed appropriately, clinicians and patients can benefit from its use within acute care settings. Although the association between usability and varying levels of EHR implementation or functionality was not determined, it is an important area for future research. Ultimately, these findings emphasize the consequence of poor usability and reinforce the need for decision makers and EHR vendors to prioritize the optimization of systems collaboratively to better support clinical practice within acute care.

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